

ASWIN KARTHIK VIJAYAKUMAR

Software Development Engineer | Full-Stack Developer | AI/GenAI

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PROFESSIONAL SUMMARY

Computer Science Engineering undergraduate (Class of 2027, CGPA 8.46/10) with strong foundations in Data Structures & Algorithms, Object-Oriented Design, and Distributed Systems. Experienced designing scalable REST APIs, cloud-native microservices, and full-stack apps using Java, Python, React.js, and SQL/NoSQL. Builder of several AI-powered platforms leveraging GenAI tools to accelerate development, with a customer-obsessed, ownership-driven mindset suited to fast-moving agile environments.

EDUCATION

SRM Institute of Science and Technology

Aug 2023 - Aug 2027

B.Tech, Computer Science and Engineering | CGPA: 8.46/10 | Chennai, India

- Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, DBMS, Computer Networks, Software Engineering, Complexity Analysis, Cloud Computing (Basics).

TECHNICAL SKILLS

- Programming Languages:** Java, Python, C, JavaScript, TypeScript (Basics)
- CS Fundamentals:** Data Structures & Algorithms, Object-Oriented Design (OOD), Distributed Systems fundamentals, Complexity Analysis, Software Development Life Cycle (SDLC), Design Patterns, Agile/CI-CD principles
- Backend & APIs:** Python Flask, FastAPI (Basics), REST API Design, Microservices (Basics), Node.js (Basics)
- Frontend:** React.js, HTML5, CSS3, JavaScript, Responsive Web Design
- Databases:** SQL, MySQL, MongoDB (Basics/NoSQL), Supabase
- Cloud & DevOps Tools:** AWS (Basics), Docker, Git/GitHub, Linux, Postman, CI/CD pipelines, Version Control
- Testing & Professional Skills:** Unit Testing (JUnit), Debugging, Problem Solving, Collaboration, Communication Skills
- AI/ML:** Generative AI / LLM Integration (Anthropic & Gemini APIs), Prompt Engineering, RAG (Basics), Machine Learning, NumPy, Pandas

EXPERIENCE

Web Developer Intern - Prodigy InfoTech

Feb 2025 - Mar 2025

- Developed responsive, scalable web applications using HTML, CSS, JavaScript, Python Flask, and SQL, following object-oriented and modular design principles.
- Designed and built REST APIs, integrating backend services with relational databases to support core application workflows.
- Improved application performance and reliability through systematic debugging, code refactoring, and peer code reviews.
- Used Git/GitHub for version control, branching, and collaborative code management in an agile team setting.

PROJECTS

BusinessOS AI - Autonomous Business Operations Platform

[GitHub](#) / [Live](#)

- Built a scalable full-stack automation platform (React.js, Python Flask, MySQL, REST APIs) with modular, cloud-ready backend microservices for secure auth and workflow automation.
- Built responsive dashboards with real-time database integration, applying clean software engineering and design-pattern best practices.

CyberShield AI - AI-Powered Cybersecurity Copilot

[GitHub](#) / [Live](#)

- Developed an AI-powered security platform (React, TypeScript, Python, Gemini API, Supabase) with scalable REST APIs for real-time threat analysis and AI-driven recommendations.

SentinelIX - SOC Dashboard Simulation Platform

[GitHub](#) / [Live](#)

- Engineered an interactive SOC simulation (React, Chart.js) with SIEM alerting, MITRE ATT&CK mapping, threat-intel IOC lookups, and an Anthropic-API-powered AI Analyst chat assistant.

Quantbot - Automated Trading & Risk Platform

[GitHub](#) / [Live](#)

- Built a Python-based multi-broker trading platform (Alpaca/OANDA adapters) with a FastAPI backend and monitoring dashboard; added an AI-powered trade-audit layer via the Anthropic API for automated review.

RESEARCH & ADDITIONAL PROJECTS

- AI-NEXUS: Co-authored an IEEE-style paper on Near-Earth Asteroid collision-risk prediction combining physics-based modeling with ML.
- NeuroSense AI: Designed an AI-powered Parkinson's-detection prototype, including UML, ER, and DFD architecture diagrams.

CERTIFICATIONS

- SAP Certified - Database Administrator (SAP HANA)
- Web Development - Internshala
- Introduction to Artificial Intelligence - IBM

ACHIEVEMENTS

- Selected for the PwC Acceleration Launchpad Program.
- Actively solving Data Structures & Algorithms problems on HackerRank, progressing through intermediate-level DAA badges.
- Built and deployed 5+ full-stack and AI-powered software applications using Python, React.js, Flask, and REST APIs.